

Exercice 4:

$$\begin{cases} u_0 = 9 \\ u_{n+1} = -\frac{1}{3} u_n \end{cases}$$

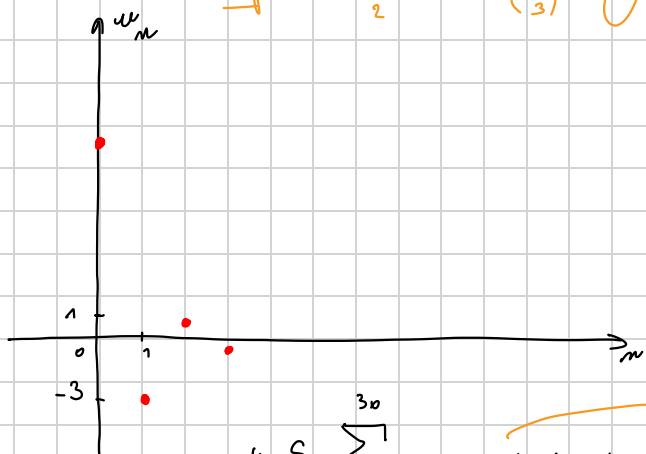
suite géométrique de
raison $q = -\frac{1}{3}$

1. $u_1 = -\frac{1}{3} u_0 = -\frac{1}{3} \times 9 = -3$

$u_2 = 1$
 $u_3 = -\frac{1}{3}$
 $u_4 = \frac{1}{9}$

2. $u_n = u_0 \times q^n = 9 \times \left(-\frac{1}{3}\right)^n$

vérif : $u_2 = 9 \times \left(-\frac{1}{3}\right)^2 = 1$



4. $S = \sum_{i=0}^{30} u_i = u_0 + u_1 + u_2 + \dots + u_{30}$ 31 termes

$S = \boxed{} \times \frac{1-q}{1-q}$
1^{er} terme $\leftarrow$ nb de termes

$S = 9 \times \frac{1 - \left(-\frac{1}{3}\right)^{31}}{1 - (-1/3)} \approx 6,75$

vérif tableau.

Ex 5 :

$$\begin{cases} u_0 = \frac{1}{3} \\ u_{n+1} = 4 u_n \end{cases}$$

1. $u_1 = \frac{4}{3} \quad u_2 = \frac{16}{3} \quad u_3 = \frac{64}{3}$

2. $u_n = u_0 q^n = \frac{1}{3} \times 4^n$

3. $S = \sum_{i=15}^{27} u_i \quad q=4$

$$S = u_{15} \times \frac{1 - q^{13}}{1 - q}$$

$$S = \frac{1}{3} \times 4^{15} \times \frac{1 - 4^{13}}{1 - 4} \approx 8 \times 10^{15}$$

Ex 6 : (u_n) suite géométrique de raison b , $b > 0$

$$u_0 = 10000$$

1. $u_n = 10000 \times b^n$

2. $u_4 = 14641$

$$b = ?$$

$$10000 \times b^4 = 14641$$

$$b^4 = 1,4641$$

$$b = \sqrt[4]{1,4641} = 1,4641^{\frac{1}{4}}$$

$$b \approx 1,1$$

Ex 7c : (u_n) suite arithmétique de raison a

$$u_3 = 5$$

$$S_4 = 15$$

$$u_0 = ?$$

$$a = ?$$